CYBER & NETWORK SECURITY

FIVE MAJOR CYBERATTACKS

Detailed article-style visuals • case-study analysis • definitions • prevention lessons

A new edition featuring Equifax, SolarWinds, Colonial Pipeline, Marriott/Starwood and MGM Resorts - five widely documented incidents with different attack methods and consequences.

Each case includes a large article-style evidence page followed by a separate analysis page with definition, timeline, impact, response, security lessons and live source links.

Editorial note: Article visuals are large, original educational reconstructions based on the cited reporting. They are not literal publisher screenshots or scans of original newspapers. The report summarises source information in its own wording and links to the full source for verification.

How to use this report

Each case starts with a full-page newspaper-style visual containing the source, headline, date, facts and a detailed summary. The following page defines the attack, explains the timeline and impact, summarises the response, and sets out practical prevention lessons.

Five cases selected

1. Equifax consumer-data breach - 22 July 2019

2. SolarWinds Orion supply-chain attack - 13 December 2020

3. Colonial Pipeline ransomware attack - 7 June 2021

4. Marriott-Starwood guest database breach - 30 November 2018

5. MGM Resorts cybersecurity incident - 12 September 2023

Key terms

Data breach: unauthorised access to or disclosure of protected information.

Supply-chain attack: compromise of a supplier, product or software-update process used by downstream organisations.

Ransomware: malicious software used to block access to systems or information, often to demand payment.

Cybersecurity incident: an event that threatens the confidentiality, integrity or availability of systems or information.

Evidence note: Company statements, regulator findings and media reporting can describe different stages of an investigation. This report flags changing estimates and does not treat unverified entry-path claims as established facts.

FEDERAL TRADE COMMISSION

CYBERSECURITY CASE 01

EQUIFAX / NEWS & OFFICIAL REPORTING

Equifax to Pay $575 Million as Part of Settlement Related to 2017 Data Breach

A preventable vulnerability and weak follow-up on patching exposed highly sensitive personal information belonging to millions of people.

BY FTC OFFICE OF PUBLIC AFFAIRS

PUBLICATION / ALERT DATE: 22 July 2019

147 MILLION
people affected in the 2017 breach

145.5 MILLION
Social Security numbers reported stolen

$575M+
announced minimum settlement in 2019

THE BREACH

In September 2017, credit-reporting company Equifax announced a major breach involving personal data. The Federal Trade Commission later said the incident affected approximately 147 million people. The exposed information included names, dates of birth, addresses, Social Security numbers and other information that could enable identity theft or fraud.

HOW ATTACKERS GOT IN

The FTC stated that Equifax had been alerted in March 2017 to a critical vulnerability affecting its ACIS database. A patch order was issued, but the company did not effectively verify that the affected systems had been updated. In July, suspicious traffic was detected; investigators found that attackers had exploited the unpatched weakness.

WHY THE DAMAGE SPREAD

The FTC's account described additional control failures: credentials were stored in plain text, database servers were not sufficiently segmented, and intrusion detection protections were inadequate for legacy databases. The breach was therefore not just a missing software update. Multiple layers of security did not prevent access from expanding to a large amount of sensitive information.

WHAT INFORMATION WAS EXPOSED

The FTC reported that at least 147 million names and dates of birth, 145.5 million Social Security numbers and 209,000 payment-card numbers and expiration dates were stolen. Personal information of this kind can be difficult or impossible for a person to change, unlike a password, so the consequences may continue long after an incident is contained.

THE REGULATORY RESPONSE

In July 2019, Equifax agreed to a settlement with the FTC, the Consumer Financial Protection Bureau and U.S. states and territories. The announced settlement was at least $575 million and could reach $700 million. It included consumer compensation, credit-monitoring support and requirements to improve the company's security programme.

THE LONG-TERM LESSON

A patch policy works only when the organisation verifies that updates were actually installed. Sensitive data also needs layers of protection: access controls, network segmentation, monitoring, protected credentials and independent checks that the controls are working. A single missed update should not grant an intruder broad access to millions of records.

RECONSTRUCTED ARTICLE VISUAL

Summarised from the cited source; not a literal screenshot or original newspaper scan.

CASE 1 / ANALYSIS

Equifax consumer-data breach

Incident / disclosure: 22 July 2019 **Case:** EQUIFAX

Definition of the attack

A data breach is an incident in which unauthorised people access or take protected information. Equifax was a vulnerability-exploitation incident: attackers used a known software weakness that had not been successfully patched, then reached data held by the company.

IMPACT	RESPONSE
Sensitive personal data associated with about 147 million people was exposed, creating risks of identity theft, fraud and long-term harm to consumers.	Equifax investigated, notified consumers, offered assistance and later agreed to a major regulatory settlement. The FTC settlement required a comprehensive information-security programme, assessments and stronger controls.
TIMELINE	**SOURCE NOTE**
Mar 2017 - FTC says Equifax received a warning about a critical vulnerability and ordered patching. **Jul 2017** - Suspicious traffic was detected; the company began investigating. **Sep 2017** - Equifax publicly announced the breach. **Jul 2019** - FTC and other regulators announced a settlement requiring relief and security improvements.	Source distinction: The FTC press release summarises allegations and settlement requirements. Its figures and description are official regulatory reporting; the case summary does not imply that every allegation is a judicial finding.

Prevention and lessons learned

- Track security patches centrally and verify completion on every affected system.
- Do not store administrator credentials in plain text.
- Segment networks and databases so one compromised service cannot reach everything.
- Monitor important systems and investigate suspicious traffic quickly.
- Test security controls and document who is responsible for each control.

Primary source:
https://www.ftc.gov/news-events/news/press-releases/2019/07/equifax-pay-575-million-part-settlement-ftc-cfpb-states-related-2017-data-breach

Additional source: https://www.ftc.gov/enforcement/refunds/equifax-data-breach-settlement

SOLARWINDS / NEWS & OFFICIAL REPORTING

Active Exploitation of SolarWinds Software

Attackers compromised a trusted software-update process, allowing malicious code to reach government agencies and private-sector organisations through legitimate updates.

BY CISA PUBLIC ALERT

PUBLICATION / ALERT DATE: 13 December 2020

MAR-JUN 2020
affected Orion releases were distributed

13 DEC 2020
CISA published an alert

SUPPLY CHAIN
trusted software updates carried the backdoor

THE ALERT

On 13 December 2020, the U.S. Cybersecurity and Infrastructure Security Agency (CISA) warned of active exploitation of SolarWinds Orion Platform software. The affected releases had been published between March and June 2020. Orion is used by organisations to monitor and manage IT environments, so it normally has trusted visibility inside networks.

HOW THE ATTACK WORKED

Rather than beginning with a separate phishing email to every victim, attackers tampered with the software supply chain. A malicious component was inserted into updates that appeared to be legitimate. Organisations that installed the affected updates could therefore introduce a backdoor while carrying out an ordinary software-maintenance task.

WHY TRUST WAS THE TARGET

Software-management tools often run with broad privileges and can communicate with many systems. Compromising a trusted vendor or build-and-release process can give attackers a path into many customers at once. The malicious update was only the beginning: the attackers then chose some environments for further activity.

A WIDER INVESTIGATION

CISA's follow-up technical alert described compromises at U.S. government agencies, critical infrastructure entities and private-sector organisations. It warned that remediation could be complex because the threat actor had remained inside some victim networks for an extended period and had used more than one possible initial access route.

CONTAINMENT

CISA issued Emergency Directive 21-01 in December 2020, directing federal civilian agencies to disconnect affected SolarWinds Orion products and take steps to mitigate the risk. FireEye/Mandiant and SolarWinds also released advisories and detection guidance. The response required organisations to investigate what had happened after the backdoor arrived, not merely uninstall an update.

THE LONG-TERM LESSON

The incident changed how organisations think about third-party risk. A vendor's trusted status does not eliminate risk. Customers need an inventory of software dependencies, protected update infrastructure, monitoring for unusual outbound communications, and an incident plan that can rapidly isolate a trusted tool without losing control of the network.

RECONSTRUCTED ARTICLE VISUAL
Summarised from the cited source; not a literal screenshot or original newspaper scan.

CASE 2 / ANALYSIS

SolarWinds Orion supply-chain attack

Incident / disclosure: 13 December 2020 **Case:** SOLARWINDS

Definition of the attack

A software supply-chain attack compromises a product, update process or supplier so that downstream customers receive malicious code through a channel they normally trust. In the SolarWinds incident, a compromised Orion update was one documented route into affected organisations.

IMPACT	RESPONSE
The campaign affected government and private-sector environments and created difficult investigations because the attackers used a trusted software channel and could conduct follow-on activity in selected networks.	CISA issued urgent guidance and an emergency directive; SolarWinds and security firms released advisories and detection measures. Affected organisations had to assess both the compromised update and potential post-compromise activity.
TIMELINE	**SOURCE NOTE**
Mar-Jun 2020 - Affected SolarWinds Orion releases were distributed. **Dec 2020** - The intrusion became public and CISA issued an alert about active exploitation. **Dec 2020 onward** - Federal agencies were directed to disconnect affected products; vendors and agencies investigated and issued mitigation guidance.	CISA confirms that affected Orion versions were released between March and June 2020 and describes broader victims and response. The report does not equate every organisation that received an update with a confirmed hands-on intrusion.

Prevention and lessons learned

- Maintain an accurate inventory of critical software and vendors.
- Secure build systems and software-signing/update processes.
- Monitor trusted management tools for unusual network behaviour.
- Use least privilege for monitoring platforms that can reach many systems.
- Prepare response plans for incidents that cross supplier and customer boundaries.

Primary source: https://content.govdelivery.com/accounts/USDHSCISA/bulletins/2b0f481

Additional source: https://content.govdelivery.com/accounts/USDHSCISA/bulletins/2b1b32d

COLONIAL PIPELINE / NEWS & OFFICIAL REPORTING

Department of Justice Seizes $2.3 Million in Cryptocurrency Paid to DarkSide

A ransomware incident at a major fuel-pipeline operator led to shutdowns of parts of the infrastructure and a national response focused on critical-service resilience.

BY DOJ OFFICE OF PUBLIC AFFAIRS

PUBLICATION / ALERT DATE: 7 June 2021

7 MAY 2021
Colonial Pipeline reported a network incident

75 BITCOIN
approximate ransom paid, per DOJ report

$2.3 MILLION
value of cryptocurrency later seized by DOJ at the time

THE INCIDENT

Colonial Pipeline disclosed a ransomware incident in May 2021. The company took portions of its infrastructure out of operation while it responded. The disruption drew national attention because the pipeline is an important part of the fuel supply system serving the eastern United States.

DARKSIDE RANSOMWARE

The FBI confirmed that the DarkSide ransomware was responsible for the compromise of Colonial Pipeline's networks. Ransomware operators typically use malicious software to disrupt access to systems and combine that disruption with a demand for payment. The intrusion can force a company to prioritise safety and continuity decisions while it investigates the scope of the compromise.

WHY A SHUTDOWN HAPPENED

The decision to stop or limit operations during a cyber incident can be a risk-management step, not proof that every industrial control system was encrypted. Operators may suspend parts of a service to protect systems, determine what is affected and prevent the incident from spreading while digital and operational teams coordinate.

RANSOM AND RECOVERY

The Department of Justice said Colonial Pipeline reported paying a demand of approximately 75 bitcoin. On 7 June 2021, DOJ announced it had seized 63.7 bitcoin, then valued at approximately $2.3 million, which it said represented proceeds traceable to the payment.

LAW ENFORCEMENT RESPONSE

The FBI confirmed DarkSide's involvement and worked with the company and government partners. The DOJ seizure showed that cryptocurrency payments may be traceable on public ledgers and that law enforcement may be able to recover some proceeds. It did not erase the disruption or guarantee that all ransom funds could be recovered.

THE CRITICAL-INFRASTRUCTURE LESSON

Essential-service operators need to plan for the possibility that corporate IT and operational technology will be affected together, even when the precise technical boundaries are different. Clear segmentation, offline backups, tested restoration and close coordination with authorities help reduce downtime and uncertainty.

RECONSTRUCTED ARTICLE VISUAL

Summarised from the cited source; not a literal screenshot or original newspaper scan.

CASE 3 / ANALYSIS

Colonial Pipeline ransomware attack

Incident / disclosure: 7 June 2021 **Case:** COLONIAL PIPELINE

Definition of the attack

Ransomware is malicious software used to block access to information or systems, often alongside a demand for payment. The Colonial Pipeline case illustrates how a ransomware compromise of a company's computer network can lead to broader operational precautions and public-service disruption.

IMPACT	RESPONSE
The company took portions of infrastructure out of operation, contributing to fuel-supply disruption and emergency response measures.	Colonial Pipeline contacted the FBI; federal agencies investigated; DOJ later seized cryptocurrency linked to part of the ransom payment. The case prompted broad focus on resilience of critical infrastructure.
TIMELINE	**SOURCE NOTE**
7 May 2021 - Colonial Pipeline experienced the network compromise and took parts of its infrastructure out of operation. **10 May 2021** - The FBI confirmed DarkSide ransomware was responsible. **7 Jun 2021** - DOJ announced the seizure of 63.7 bitcoin linked to the ransom payment.	DOJ and FBI sources support the timeline, DarkSide attribution and ransom-recovery figures. The page avoids assuming that every operational disruption came directly from encryption of industrial control systems.

Prevention and lessons learned

- Separate corporate IT from operational technology wherever possible.
- Keep tested, isolated backups and documented recovery plans.
- Practice safe shutdown and restart procedures for critical services.
- Report incidents early and coordinate with law enforcement and regulators.
- Prepare communications plans for customers, employees and public authorities.

Primary source: https://www.justice.gov/archives/opa/pr/department-justice-seizes-23-million-cryptocurrency-paid-ransomware-extortionists-darkside

Additional source: https://www.fbi.gov/news/press-releases/fbi-statement-on-compromise-of-colonial-pipeline-networks

MARRIOTT / STARWOOD / NEWS & OFFICIAL REPORTING

Marriott Announces Starwood Guest Reservation Database Security Incident

Unauthorised access to the Starwood reservation database continued for years before the hotel group identified and disclosed the breach.

BY MARRIOTT INTERNATIONAL STATEMENT

PUBLICATION / ALERT DATE: 30 November 2018

SINCE 2014
unauthorised access was later traced back to

30 NOV 2018
Marriott publicly announced the incident

~383 MILLION
upper limit of records in later SEC filing analysis

THE DISCLOSURE

On 30 November 2018, Marriott announced a security incident involving the Starwood guest reservation database. During an investigation, the company determined that an unauthorised party had accessed the Starwood network since 2014 and had copied information from the reservation database.

WHAT THE DATABASE CONTAINED

The records included a range of guest details depending on the individual: names, mailing addresses, phone numbers, email addresses, passport numbers, reservation details, arrival and departure information, loyalty-programme information and other preferences. The mix of fields varied, so not every affected record necessarily contained every category of information.

A LONG DWELL TIME

The case showed how a breach can remain active for years before it is detected and fully understood. Marriott received an alert in September 2018 about an attempt to access the database and later determined that information had been copied. The acquisition of Starwood did not itself create the intrusion, but the inherited system became part of Marriott's investigation and response.

HOW MANY RECORDS?

Marriott's initial announcement said the database may have contained information relating to up to approximately 500 million guests. A later SEC filing described an upper limit of approximately 383 million guest records based on the company's analysis at that time. This difference illustrates why a case study should identify when an estimate was made instead of treating early figures as final.

WHAT MADE IT SERIOUS

The combination of contact information, travel details and passport or payment information can increase the risk of phishing, identity theft and targeted fraud. Even if a password is changed, passport numbers and past reservation records cannot simply be replaced in the same way.

SECURITY AND GOVERNANCE LESSON

Large organisations inherit systems, data stores and technical debt through acquisitions. A merger or acquisition should therefore include a careful security review, log retention, access audits and a defined plan for migrating or isolating legacy databases. Monitoring must be effective enough to identify suspicious access before it persists for years.

RECONSTRUCTED ARTICLE VISUAL

Summarised from the cited source; not a literal screenshot or original newspaper scan.

CASE 4 / ANALYSIS

Marriott-Starwood guest database breach

Incident / disclosure: 30 November 2018 **Case:** MARRIOTT / STARWOOD

Definition of the attack

A database breach occurs when an unauthorised party accesses or copies protected records. The Marriott-Starwood incident involved persistent access to a guest reservation database and the copying of personal information over a period of years.

IMPACT	RESPONSE
Guest records containing personal, travel, loyalty and in some cases passport or payment-card information were involved, creating substantial privacy, fraud and regulatory concerns.	Marriott engaged security experts, investigated the database, notified relevant parties and made public disclosures. Later filings updated the record-count estimate and discussed what information was involved.
TIMELINE	**SOURCE NOTE**
2014 - Marriott later said unauthorised access to the Starwood network dated back to this year. **8 Sep 2018** - Marriott received an alert about an attempt to access the guest reservation database. **30 Nov 2018** - The company disclosed the incident publicly. **2019 filing** - A later SEC filing described an upper limit of approximately 383 million records based on analysis at that time.	Marriott's 2018 release gives the initial disclosure and estimate; the cited SEC filing later reported approximately 383 million as the upper limit of records based on analysis at that time.

Prevention and lessons learned

- Audit security before and after acquiring another company's systems.
- Keep detailed logs and alert on unusual database access or large data exports.
- Retire or isolate legacy systems that cannot meet current security standards.
- Protect passport and payment information with strong controls and careful key management.
- Communicate evolving estimates transparently and correct earlier figures when evidence changes.

Primary source: https://marriott.gcs-web.com/news-releases/news-release-details/marriott-announces-starwood-guest-reservation-database-security

Additional source: https://www.sec.gov/Archives/edgar/data/1048286/000162828019002337/mar-q42018x10k.htm

MGM RESORTS / NEWS & OFFICIAL REPORTING

MGM Resorts International Statement on Cybersecurity Issue

A cybersecurity incident disrupted hotel and casino operations, forced systems offline and resulted in some personal information being obtained.

BY MGM RESORTS INTERNATIONAL

PUBLICATION / ALERT DATE: 12 September 2023

SEP 2023
incident identified by MGM Resorts

~$100 MILLION
estimated negative impact on adjusted property EBITDAR in Q3

SOME GUEST DATA
personal information of some customers was obtained

THE PUBLIC STATEMENT

On 12 September 2023, MGM Resorts said it had identified a cybersecurity issue affecting certain U.S. systems. The company began an investigation with external cybersecurity experts, notified law enforcement and shut down some systems to protect information and contain the incident.

VISIBLE OPERATIONAL DISRUPTION

MGM properties experienced interruptions involving guest-facing systems and hotel operations. When reservation, check-in, room-key, payment or internal applications become unavailable, employees may have to switch to manual processes. The disruption can affect guests even if a particular payment system or customer database is not shown to have been accessed.

THE INFORMATION IMPACT

In its later annual filing, MGM said the attackers obtained certain personal information for some customers, including names, contact details, gender, date of birth and driver's licence numbers. For a limited number of customers, Social Security numbers and passport numbers were also obtained. MGM said it did not believe customer passwords, bank account numbers or payment-card information had been obtained.

BUSINESS COST

In an SEC filing, MGM estimated that the September disruption would have a negative impact of approximately $100 million on adjusted property EBITDAR for its Las Vegas Strip resorts and regional operations. The company also reported less than $10 million in one-time third-quarter expenses related to the incident, while noting that the full scope of costs had not been determined.

WHAT IS CONFIRMED

MGM's public filings confirm unauthorised access, shutdown of some systems, operational disruption and exposure of certain personal data. Public reporting has discussed social engineering as a possible route, but the company filings cited here do not provide a complete technical account of the initial access path. The report therefore separates documented impact from public theories about the entry method.

WHY RESILIENCE MATTERS

The incident illustrated that security controls should help a company recover as well as block an attacker. Organisations need incident-response playbooks, access controls, reliable backups, fallback procedures for guest operations, and careful communication with staff and customers during service interruptions.

RECONSTRUCTED ARTICLE VISUAL

Summarised from the cited source; not a literal screenshot or original newspaper scan.

CASE 5 / ANALYSIS

MGM Resorts cybersecurity incident

Incident / disclosure: 12 September 2023 **Case:** MGM RESORTS

Definition of the attack

A cybersecurity incident is an event that compromises the confidentiality, integrity or availability of systems or information. The MGM case involved unauthorised access to company systems, operational disruption and exposure of some personal information.

IMPACT	RESPONSE
Operations at domestic properties were disrupted; some guest information was obtained; MGM estimated around $100 million in negative adjusted property EBITDAR impact for the third quarter.	MGM worked with external cybersecurity experts, notified law enforcement and shut down certain systems. Its later filing described recovery and the types of personal information involved.
TIMELINE	**SOURCE NOTE**
12 Sep 2023 - MGM publicly disclosed a cybersecurity issue and said it had shut down certain systems. **Sep 2023** - Guest-facing systems and some hotel operations were disrupted while investigations and recovery continued. **SEC filings** - MGM disclosed personal information affected for some customers and estimated the incident's financial impact.	MGM's SEC filings are used for the documented impact and data types. Media reports have discussed potential social-engineering methods, but this report does not claim that the cited company filings prove a specific initial-access method.

Prevention and lessons learned

- Use strong identity checks for help-desk requests and privileged-access recovery.
- Limit permissions between corporate and property-operation systems.
- Prepare manual fallback processes for check-in and other guest-facing services.
- Maintain a rapid incident-response plan that covers operational recovery and communication.
- State clearly which data is known to be affected and what remains under investigation.

Primary source: https://www.sec.gov/Archives/edgar/data/789570/000119312523233855/d502352dex991.htm

Additional source: https://www.sec.gov/Archives/edgar/data/789570/000078957024000005/mgm-20231231.htm

Lessons across all five incidents

Different sectors, different attack paths - but recurring security weaknesses appear across the cases.

1. Patch management is a process, not a reminder

Equifax shows why a patch command is not enough: teams need to verify that systems were updated and that unresolved exceptions are tracked. Critical vulnerabilities should have clear owners, deadlines and independent checks.

2. Trust must be verified throughout the software supply chain

SolarWinds shows that a trusted software update can become an attack channel. Organisations should inventory dependencies, protect build and release systems, and watch for unusual behaviour from trusted management tools.

3. Critical services need recovery plans

Colonial Pipeline and MGM illustrate that a cyber incident can disrupt physical-world services and customer operations. Recovery plans should include safe shutdown, tested restoration, manual fallbacks and clear communication.

4. Protect sensitive records for their full life cycle

Equifax and Marriott show why personal information needs strict access, secure storage, database monitoring and retention limits. Long-lived personal details create lasting risks when exposed.

5. Separate confirmed findings from early estimates

A report should identify whether a statement comes from an official filing, regulator, company announcement or early media report. Estimates can change as investigations identify duplicates, affected accounts and the full scope of an incident.

Sources and verification

The links below are the primary source material used for the case pages, with an additional official filing or update where useful. Open the linked sources to consult the full details and wording.

1. Equifax consumer-data breach

Primary source:
https://www.ftc.gov/news-events/news/press-releases/2019/07/equifax-pay-575-million-part-settlement-ftc-cfpb-states-related-2017-data-breach

Additional source: https://www.ftc.gov/enforcement/refunds/equifax-data-breach-settlement

2. SolarWinds Orion supply-chain attack

Primary source: https://content.govdelivery.com/accounts/USDHSCISA/bulletins/2b0f481

Additional source: https://content.govdelivery.com/accounts/USDHSCISA/bulletins/2b1b32d

3. Colonial Pipeline ransomware attack

Primary source: https://www.justice.gov/archives/opa/pr/department-justice-seizes-23-million-cryptocurrency-paid-ransomware-extortionists-darkside

Additional source: https://www.fbi.gov/news/press-releases/fbi-statement-on-compromise-of-colonial-pipeline-networks

4. Marriott-Starwood guest database breach

Primary source: https://marriott.gcs-web.com/news-releases/news-release-details/marriott-announces-starwood-guest-reservation-database-security

Additional source: https://www.sec.gov/Archives/edgar/data/1048286/000162828019002337/mar-q42018x10k.htm

5. MGM Resorts cybersecurity incident

Primary source: https://www.sec.gov/Archives/edgar/data/789570/000119312523233855/d502352dex991.htm

Additional source: https://www.sec.gov/Archives/edgar/data/789570/000078957024000005/mgm-20231231.htm

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